Ram Calculus for the Anti-de Sitter Penrose Construction

Saturating Stress, Finite-Height Area Bounds,
Convex-Dual Flux Certificates, and Static Deficits

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Abstract

We investigate the conserved-stress and static-deficit mechanisms in the October 5, 2026 manuscript on the nonmaximal anti-de Sitter Penrose inequality. The principal local results are an explicit Legendre dual of its saturating constitutive law, a primal–dual certificate controlling flux error despite loss of uniform ellipticity in the potential equation, and a quantitative $O(D^{-1})$ enclosure-area approximation at fixed prepared geometry. The last estimate combines the source's height-independent distant-end barrier with its full-cut coarea argument. We also derive finite-radius four-flux error bounds, a timelike mass certificate, and a residual-controlled static-deficit test. Compatible Ram moment spaces and exact polynomial enclosures preserve the actual divergence, metric, and trace constraints; the corpus's superlinear coercivity and linear hypercircle theorems cannot simply be substituted for this linear-growth nonlinear problem. Each proved refinement is distinguished from imported existence, compactness, end-bending, and rigidity statements. No independent proof of the complete nonmaximal Penrose theorem, quantitative geometric rigidity, or computational speedup is claimed. Contribution assessment: C3/5, explicit nonlinear flux certification and a finite-height rate refinement with complete restricted proofs, but no independently certified global Penrose or rigidity theorem.

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1 Scope and Source Status

The assigned source is OpenAI, *The nonmaximal anti-de Sitter Penrose Inequality and original-data rigidity*, October 5, 2026, at the pinned repository commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its source directory is the hyphenated title followed by `-October-5-2026`, under `_sources/openai-math/preprints/`. Literal labels below refer to files in its `build/sections/` directory. The main assembly, introduction, history, setup, the entire 1,799-line `03-stress-bending.tex`, and `11-static-rigidity.tex` were read directly. Selected physical-equation, end-estimate, and final-invariant passages of module 07 were read; the independent bounded audit examines modules 06 and 13. This is a detailed investigation of specified mechanisms, not a claim that every proof in all modules 04–12 has been independently checked. In particular, the full coupled existence and original-data rigidity chains remain dependencies.

The distinction is essential for a new preprint containing a major general theorem. The public repository's catalogue describes the asserted result, but a catalogue entry is not outside confirmation. The present analysis establishes its own propositions under explicitly stated hypotheses. A source lemma used as an input is always identified. Algebraic tests supplement proofs; they do not establish geometric compactness or correctness of an entire manuscript.

The applicability rating L4 concerns relevance. The contribution rating C3 concerns the restricted additions actually proved here. Neither means that the unrestricted geometric problem has been independently solved by Ram Calculus. No confidential corpus text was transmitted to a public literature search.

2 The Exact Geometric Problem

Work in spatial dimension three with cosmological constant $\Lambda = -3$ and $G = c = 1$. The data (Ω, g, K) are smooth, oriented, connected, complete with their nonempty compact boundary S included, and have exactly one spherical asymptotically hyperbolic end. Put

$$b = \frac{\mathrm{d}r^2}{1+r^2} + r^2\sigma, \quad 16\pi\mu = R_g + 6 + (\mathrm{tr}_g\, K)^2 - |K|_g^2, \quad 8\pi J = \mathrm{div}_g(K - (\mathrm{tr}_g\, K)g). \tag{2.1}$$

The source assumes $\mu \geq |J|_g$, physical weighted decay $g - b \in C^{2,\alpha}_\tau$, $K \in C^{1,\alpha}_\tau$, $3/2 < \tau < 3$, and

$$\int_\Omega \sqrt{1+r^2}\,(\mu + |J| + |K|^2)\,\mathrm{d}V_g < \infty. \tag{2.2}$$

The boundary condition is the *same future sign* $\theta_+ = H + \mathrm{tr}_{TS}\, K \leq 0$ on every component, with normal pointing into the designated exterior. It is not the arbitrary mixture of future and past signs appearing in some different charged manuscripts.

A full cut is the entire intrinsic compact boundary of a connected closed exterior submanifold containing the distant end. All components and contact portions count. Taking the whole exterior gives the original boundary, not an empty zero-area cut. Write

$$A = \inf_{\Gamma \text{ full cut}} \mathrm{Area}_g(\Gamma), \qquad r_A = \sqrt{A/(4\pi)}, \qquad B(A) = \frac{1}{2}(r_A + r_A^3). \tag{2.3}$$

The numerical source assertion is $m_{\mathrm{AH}} \geq B(A)$, with the mass convention below. Rigidity additionally imposes a connected outermost marginally future-trapped boundary that is outer-area-minimizing. The

source then identifies the *original* (g, K) as an appropriate spacelike slice in Schwarzschild–anti-de Sitter with the matching mass and full horizon section. Equality for an arbitrary nonoutermost boundary is not classified by that statement.

2.1 Metric four-flux and its normalization

Let $V_0 = \sqrt{1+r^2}$ and $V_i = r\omega_i$, $1 \le i \le 3$. They satisfy $\mathrm{Hess}_b V_a = V_a b$, $\Delta_b V_a = 3V_a$. For $e = g - b$ set

$$U_b(V, e) = V(\mathrm{div}_b\, e - \mathrm{d}\,\mathrm{tr}_b\, e) + (\mathrm{tr}_b\, e)\mathrm{d}V - e(\nabla_b V, \cdot), \tag{2.4}$$

$$p_a = \frac{1}{16\pi} \lim_{R\to\infty} \int_{r=R} U_b(V_a, e)(\nu_b)\, \mathrm{d}A_b, \qquad m_{\mathrm{AH}} = \sqrt{p_0^2 - |\mathbf{p}|^2}. \tag{2.5}$$

The source requires $p_0 > |\mathbf{p}|$. This is the *metric* hyperbolic four-flux; it is not the ADM energy–momentum formula for an asymptotically Euclidean end. A background hyperbolic spatial isometry balances a timelike four-vector. An arbitrary change of spacetime slice is not the same operation.

The model $F_M(r) = 1 + r^2 - 2M/r$, $g_M = F_M^{-1}\mathrm{d}r^2 + r^2\sigma$ has mass $(M, 0, 0, 0)$. Its horizon radius satisfies $M = (r_h + r_h^3)/2$. Thus the cubic term in B is essential; replacing the target by only $\sqrt{A/(16\pi)}$ loses the cosmological contribution.

3 Source Mechanisms and the Ram Transfer

3.1 Dependencies kept separate

The source's route is: conformal strict preparation; a nonlinear bounded-flux potential and conserved stress; conversion of a prepared hyperbolic end to a flat one with controlled mass cost; an internally proved flat coupled deformation and the classical Riemannian endpoint; a different weighted hyperbolic coupled deformation removing general K; four-flux passage and spatial Lorentz balance. Equality then uses an original-data constraint adjoint, causal Killing data, exclusion of a null set, a static quotient, conformal end improvement, a finite-domain static deficit, and original-slice recovery.

Only the stress/flux part is needed for Theorems 5.1 and 6.2. The static-deficit refinement uses a *given* smooth static system. Neither requires believing that every preceding global source construction succeeds. Conversely, none replaces the missing independent audit of those global constructions.

3.2 Actual supporting corpus proofs

The following precise parts were consulted, not just their titles. Their established classical ingredients are not claimed as new mathematics.

Corpus source	Exact transfer and limitation
TR250, `TR250-Compatible-Variational-Ram-Panels.tex`, Sections 2–5	`eq:moments`, `thm:transfer`, `eq:piola`, `thm:gamma`, `thm:hypercircle`. Compatible moment changes preserve the physical complex and fluxes. Gamma convergence requires superlinear p-growth; the hypercircle identity is quadratic and uniformly elliptic. Neither theorem applies unchanged to the saturating energy below.

Corpus source	Exact transfer and limitation
Astra IE/PDE TR8, `TR8-lin-non-lin.tex`, lines 81–199	`thm:coercive`, `eq:operator-domain`, `thm:green-equivalence`. A true boundary-domain inverse precedes a Green kernel. The weak inverse does not itself supply an H^2 estimate. It applies directly to the smooth bounded static problem after trace lifting.
RamCalBooks, volume 16, `chapters/ch05d-clearance.tex`, lines 62–145	`v16c05:conversionproof`, `v16c05:hull`, exact subdivision. Bernstein coefficients certify entire polynomial panels, not only samples. We use this for finite flux feasibility and retain separate metric and tail errors.

The first file is in the TR250 subdirectory of the user's core TRs collection; the second is in `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/`; the third is in `RamCalBooks/volumes/volume16/`. Exact read extents and file hashes are retained in the local evidence record. These proofs support a physically constrained representation and certificate, not a universal geometric existence theorem. Related corpus maps were used for routing; a map is not proof evidence.

4 The Saturating Stress Equation

On a fixed smooth, connected, complete prepared exterior (X, g) with one end and nonempty compact smooth boundary T satisfying $H_T \le 0$ toward the exterior, source `thm:str-stress-dirichlet` constructs, for every $D > 0$, a bounded weak solution f_D with trace D on T and limit zero at infinity. In the one-sign application an added product collar supplies this mean-curvature hypothesis; it is not inferred merely from $H + \operatorname{tr}_T K \le 0$. The equation is

$$\operatorname{div}_g \mathcal{A}(\nabla f_D) = 0, \qquad \mathcal{A}(p) = t(|p|)\frac{p}{|p|}, \quad \mathcal{A}(0) = 0, \quad s^2 = \frac{t(s)}{1 - t(s)^{3/2}}. \tag{4.1}$$

Here $0 \le t < 1$ and t is strictly increasing. Its logarithmic derivative is

$$b(s) = \frac{st'(s)}{t(s)} = \frac{4(1 - t^{3/2})}{2 + t^{3/2}}. \tag{4.2}$$

At zero, $t = s^2 - s^5 + O(s^8)$; at infinity, $t \to 1$ and $b(s)s^2 \to 4/3$. The radial and transverse eigenvalues of $D\mathcal{A}(p)$ are $t'(s)$ and $t(s)/s$. They approach zero at both ends of the gradient range. There is no globally uniform elliptic constant.

The source obtains finite-D gradient bounds before extending a regularized law to a global three-Laplace structure. The extension is a tool for local regularity *after* the gradient bound; it is not an existence proof for the original saturating law. It then takes regularization and outer-domain limits, uses strict monotonicity for uniqueness, and proves an end expansion

$$f_D = zC_D(\omega) + O(z^2), \quad z = r^{-1}, \quad C_D > 0, \qquad \mathcal{F}_D = \int_{S^2} C_D^2 \, \mathrm{d}\omega. \tag{4.3}$$

Prepared metrics have

$$g = z^{-2}\bar{g}, \quad \bar{g} = e^{4v}\left(\frac{\mathrm{d}z^2}{1 + z^2} + \sigma\right), \quad v = z^3 v_3 - dz^{3+\delta} + O_\infty(z^4), \quad 0 < \delta < 1. \tag{4.4}$$

The compact metric is generally only finitely regular in its normal variable; arbitrary angular smoothness does not justify arbitrary normal differentiation.

4.1 Why the compactified equation is special to three dimensions

Write $a(s) = t(s)/s^2$ and $q = |\mathrm{d}f|_{\bar{g}}$. The physical gradient has magnitude zq and vector $z^2\nabla_{\bar{g}}f$. Its flux is $z^3 a(zq)q\nabla_{\bar{g}}f$. The volume density is $z^{-3}\mathrm{d}V_{\bar{g}}$. Thus the exact transformed equation is

$$\mathrm{div}_{\bar{g}}(a(zq)q\nabla_{\bar{g}}f) = 0. \tag{4.5}$$

This is a cancellation of physical weights, not a componentwise coordinate substitution. A zero Dirichlet trace at $z = 0$ does not mean zero normal flux there. In dimension n, the corresponding density contains z^{3-n}, so the same unweighted compact equation cannot be asserted without a new constitutive choice or weighted analysis.

4.2 Reproducing the conserved stress

Put $k = \sqrt{1 - t^{3/2}}$ and $W = (3 + k^2)/(2k)$. Direct differentiation yields $W' = 3t/2$ and $W - sW' = 2k$. For $e = \nabla f/|\nabla f|$,

$$\mathcal{P} = Wg - sW'e \otimes e, \qquad L = \mathcal{P} - \tfrac{1}{2}(\mathrm{tr}\,\mathcal{P})g. \tag{4.6}$$

Its eigenvalues are $\mathcal{P}_{ee} = 2k$, $\mathcal{P}_{\perp} = W$, and

$$L_{ee} = -\frac{3 - k^2}{2k}, \qquad L_{\perp} = -k, \qquad (\mathrm{tr}\,L)^2 - |L|^2 = 6. \tag{4.7}$$

The continuous critical-point values are $\mathcal{P} = 2g$, $L = -g$. Convex additive variations of $\int W(|\mathrm{d}f|)$ give local minimality; inner variations then give $\mathrm{div}\,\mathcal{P} = 0$ distributionally. A direct pointwise differentiation only off the critical set would not by itself prove the distributional statement across that set.

If $K - (\mathrm{tr}\,K)g \geq 0$, then $\mathrm{tr}\,K \leq 0$ and $\mathrm{tr}_{e^\perp} K \leq 0$, so $\mathcal{P} : K \leq 0$. Hence

$$R_g + (\mathrm{tr}(K + L))^2 - |K + L|^2 = R_g + 6 + (\mathrm{tr}\,K)^2 - |K|^2 - 2\mathcal{P} : K \tag{4.8}$$

does not reduce the dominant-energy margin, while the momentum is unchanged. $L < 0$ improves the future expansion. This is an exact mechanism for the one-sign case. General K requires the separate hyperbolic deformation; it cannot be removed by dropping the sign hypothesis.

5 An Explicit Convex-Dual Reformulation

Define $\Phi(s) = \int_0^s t(a)\,\mathrm{d}a$ and $F(p) = \Phi(|p|)$ in each metric tangent space. The original equation is the Euler equation for this strictly convex radial energy, with prescribed traces. Its growth is cubic at zero and linear at infinity.

Theorem 5.1 (Exact dual and uniform dual convexity)**.** *Let $u = \sqrt{t(s)} \in [0, 1)$ and $k = \sqrt{1 - u^3}$. Then*

$$s = u/k, \qquad \Phi(s) = \frac{4 - u^3}{3k} - \frac{4}{3}, \tag{5.1}$$

$$F^*(q) = \Psi(|q|), \qquad \Psi(v) = \begin{cases} \frac{4}{3}(1 - \sqrt{1 - v^{3/2}}), & 0 \leq v \leq 1, \\ +\infty, & v > 1. \end{cases} \tag{5.2}$$

On the closed unit ball, F^* is strongly convex with constant $1/2$: for $|q_*| < 1$,

$$F^*(q) - F^*(q_*) - \nabla F^*(q_*) \cdot (q - q_*) \geq \frac{1}{4}|q - q_*|^2. \tag{5.3}$$

The primal energy has no global positive quadratic strong-convexity constant.

Proof. The defining law gives $s = u/k$. The displayed formula for Φ has value zero at $u = 0$ and derivative $u^2\,\mathrm{d}s/\mathrm{d}u$, proving (5.1). Radial symmetry makes the optimizing primal vector parallel to q; for $0 < v < 1$, stationarity is $t(s) = v$. Substituting $u^2 = v$ gives

$$sv - \Phi(s) = u^3/k - (4 - u^3)/(3k) + 4/3 = \tfrac{4}{3}(1 - k).$$

Continuity and monotone passage to $s = \infty$ give the value $4/3$ at $v = 1$. Since $\Phi(s)/s \to 1$, the supremum is infinite for $v > 1$.

For $0 < v < 1$, $\Psi'(v) = \sqrt{v}/\sqrt{1 - v^{3/2}}$. The Hessian eigenvalues of the radial dual are

$$\frac{\Psi'(v)}{v} = \frac{1}{\sqrt{v}\sqrt{1 - v^{3/2}}} \geq 1, \tag{5.4}$$

$$\Psi''(v) = \frac{1}{2\sqrt{v}\sqrt{1 - v^{3/2}}} + \frac{3v}{4(1 - v^{3/2})^{3/2}} \geq \frac{1}{2}. \tag{5.5}$$

Integrate the Hessian along line segments. A segment passing through zero is handled by splitting and continuity of the first derivative; a unit-sphere endpoint follows by taking an interior limit. This proves (5.3). In contrast, $\Phi(s) = s^3/3 + O(s^6)$ at zero, so $\Phi(s)/s^2 \to 0$. A positive global quadratic modulus is impossible. □

The dual removes numerical inversion of the implicit t law from the energy evaluation. It also places the natural feasibility condition $|q|_g \leq 1$ directly in the optimization. The cost is a singular dual derivative near $|q| = 1$; unconstrained smooth Newton iteration cannot cross that boundary.

5.1 A primal–dual certificate on an actual domain

Let U be a bounded smooth Riemannian domain. Assume $u \in W^{1,\infty}(U)$ solves (4.1) with given Dirichlet trace b, and put $q_* = \mathcal{A}(\nabla u)$. Let $v \in W^{1,\infty}(U)$ have the same trace, and let $q \in H(\mathrm{div}, U)$ satisfy $\mathrm{div}\, q = 0$ and $|q| \leq 1$ a.e. Define

$$\mathcal{G}(v, q) = \int_U \{F(\nabla v) + F^*(q) - q \cdot \nabla v\}\,\mathrm{d}V. \tag{5.6}$$

All terms are integrable on this bounded domain. Write $B_H(a, a_*) = H(a) - H(a_*) - \nabla H(a_*) \cdot (a - a_*)$.

Theorem 5.2 (Source-specific nonlinear flux certificate)**.** *Under these hypotheses,*

$$\mathcal{G}(v, q) = \int_U B_F(\nabla v, \nabla u) + B_{F^*}(q, q_*)\,\mathrm{d}V, \qquad \|q - q_*\|_{L^2(U)} \leq 2\sqrt{\mathcal{G}(v, q)}. \tag{5.7}$$

This bound needs no positive lower bound on $|\nabla u|$ and no uniform primal ellipticity constant.

Proof. Legendre equality gives $F(\nabla u) + F^*(q_*) = q_* \cdot \nabla u$, $\nabla F(\nabla u) = q_*$ and $\nabla F^*(q_*) = \nabla u$. Expanding the two Bregman terms shows that their sum differs from the gap integrand by $(q - q_*) \cdot \nabla(v - u)$. Its integral is zero because both fluxes have zero distributional divergence and $v - u$ has zero trace. Both Bregman terms are nonnegative; (5.3) proves the estimate. □

Equivalently, $\int F(\nabla v)$ is a primal upper bound and $\langle q \cdot n, b\rangle - \int F^*(q)$ is a dual lower bound on the minimum energy, with their difference equal to $\mathcal{G}$. Normal traces are weak traces, not sampled nodal values. In contrast to a quadratic hypercircle, the primal term measures a Bregman error; it does not give an unqualified H^1 error estimate. On an exterior the identity also holds when the displayed

integrals converge and the extra outer boundary term tends to zero. Sufficient conditions include the source's $u = O(r^{-1})$, $q_* = O(r^{-2})$, analogous decay for q, and a trial potential v that itself is compactly supported and has the required inner trace. More generally, $v - u = O(r^{-1})$ gives the vanishing outer term under those flux bounds, provided the energy and Bregman integrals converge. Merely admitting a compactly supported trace lifting does not constrain v at infinity: $v \equiv D$, $q = 0$ has zero gap but a nonzero missing outer boundary term when compared with $u = f_D$. These conditions must be checked rather than inferred from a bounded-domain discretization.

5.2 Restricted inverse-gradient estimates

On a compact patch where the entire segment between two gradients stays in $\{m \le |p| \le M\}$ for some $m > 0$, the smallest Hessian eigenvalue of F has a positive lower bound $\alpha_{m,M}$. The same argument controls the corresponding squared gradient error by $2\mathcal{G}/\alpha_{m,M}$. Merely knowing both endpoint magnitudes exceed m is insufficient: opposing gradients have a segment passing through zero. Near critical points the flux estimate, rather than this restricted gradient estimate, is the appropriate certificate.

6 Quantitative Flux Saturation

Source `lem:str-flux-saturation` proves

$$\int_X t(|\mathrm{d}f_D|)|\mathrm{d}f_D|\,\mathrm{d}V = D\mathcal{F}_D, \qquad 0 \le \mathcal{F}_D \le A, \qquad \mathcal{F}_D \longrightarrow A. \tag{6.1}$$

The area upper bound uses the divergence-free field $X_D = \mathcal{A}(\nabla f_D)$ with $|X_D| < 1$, whose oriented flux across every full cut is $\mathcal{F}_D$. The energy identity follows by integrating $\mathrm{div}(f_D X_D)$; its outer boundary term vanishes using (4.3).

For coarea, an almost-everywhere level set of f_D is a finite-perimeter set. Filling bounded complementary components can only reduce perimeter and leaves a full boundary separating every component of T from the end. Smooth approximation with the inner collar and distant exterior held fixed gives level perimeter at least A. No false smooth Sard theorem for $C^{1,\theta}$ functions is necessary.

The source obtains $(1-\lambda)A - C_0 V_\lambda/D \le \mathcal{F}_D$ and then sends $D \to \infty$, $\lambda \to 0$. A different source lemma, `lem:str-uniform-end`, supplies a stronger ingredient for a *fixed* geometry: a pole barrier independent of $D \ge 1$. In the coordinate $\eta = \operatorname{arsinh} r$, it is

$$w(\eta) = \frac{a_0 e^{-\xi(\eta-\eta_0)}}{\eta - \eta_0}, \qquad 0 < \xi < 1. \tag{6.2}$$

The source chooses a_0, η_0 so that $f_D \le w$ beyond its pole. Fix a sphere strictly outside the pole. Then $f_D \le H_0$ outside that sphere for all $D \ge 1$, and the enclosed compact part has finite volume V_0.

Lemma 6.1 (A universal saturation-loss bound)**.** *For every $s \ge 0$, $s(1 - t(s)) \le 1/2$.*

Proof. Writing $t = t(s)$ gives

$$[s(1-t)]^2 = \frac{t(1-t)^2}{1-t^{3/2}} \le t(1-t) \le \frac{1}{4},$$

because $t^{3/2} \le t$ on $[0,1]$. All quantities are nonnegative and the endpoint limits agree. □

Theorem 6.2 (Finite-height full-cut area rate)**.** *Assume the fixed-height stress solutions, energy/coarea statements above, and a fixed-geometry uniform end bound $f_D \le H_0$ outside a compact region of volume at most V_0 for every $D \ge 1$. For $D > \max(1, H_0)$,*

$$0 \le A - \mathcal{F}_D \le \frac{H_0 A + V_0/2}{D} \le \frac{H_0 A_{\mathrm{cut}} + V_0/2}{D}, \tag{6.3}$$

where A_{cut} is the area of any known smooth full cut. Thus, with $C = H_0 A_{\mathrm{cut}} + V_0/2$,

$$\mathcal{F}_D \le A \le \min\{A_{\mathrm{cut}}, \mathcal{F}_D + C/D\}. \tag{6.4}$$

Proof. The set $E_D = \{f_D > H_0\}$ is contained in the fixed compact region, hence has volume at most V_0. The nonnegative energy integrand and Lemma 6.1 give

$$D\mathcal{F}_D \ge \int_{E_D} |\mathrm{d}f_D|\,\mathrm{d}V - \tfrac{1}{2}V_0 \ge \int_{H_0}^{D} A\,\mathrm{d}a - \tfrac{1}{2}V_0 = (D - H_0)A - \tfrac{1}{2}V_0.$$

Rearranging proves the first upper bound; $A \le A_{\mathrm{cut}}$ proves the second. The lower bound and enclosure follow from (6.1). □

This theorem quantitatively combines two source arguments. It does not assume a smooth limiting stress at $D = \infty$, and it does not give constants uniform under a collapsing added collar, changing preparation index, or arbitrary families of initial data. The constants can be very large. For a genuine numerical certificate one must enclose $H_0, V_0, A_{\mathrm{cut}}$ and the flux of an actual solution, not just set them to convenient numbers.

6.1 A coarse bound requiring less end information

If only $f_D \le cDr^{-\xi}$ is known on an AH end, $0 < \xi < 1$, then $\mathrm{Vol}\{f_D > \lambda D\} \le C\lambda^{-2/\xi}$. The exponent is $2/\xi$, not $3/\xi$, since the AH radial volume is $O(r\,\mathrm{d}r)$ in this coordinate. Choosing $\lambda = D^{-1/(1+2/\xi)}$ in the source estimate gives

$$A - \mathcal{F}_D \le (A + C/2)D^{-\xi/(\xi+2)} \qquad (D \ge 1), \tag{6.5}$$

after enlarging C to cover the compact part. This weaker rate is useful when a height-independent pole barrier has not been established. It must not be used to claim the sharper rate without that additional input.

6.2 What happens to the cubic cosmological term

Hölder's inequality on S^2 gives $(4\pi)^{-1}\int_{S^2} C_D^3 \ge (\mathcal{F}_D/(4\pi))^{3/2}$. Conditional on the source's flat reduction and bending, its one-sign estimate at fixed prepared geometry is

$$p_0(g) \ge \sqrt{A/(16\pi)} + \frac{1}{2}(\mathcal{F}_D/(4\pi))^{3/2}. \tag{6.6}$$

For $0 \le \mathcal{F}_D \le A \le A_{\mathrm{cut}}$, the mean-value theorem gives

$$0 \le \frac{A^{3/2} - \mathcal{F}_D^{3/2}}{2(4\pi)^{3/2}} \le \frac{3\sqrt{A_{\mathrm{cut}}}}{4(4\pi)^{3/2}}\frac{C}{D}. \tag{6.7}$$

This is an explicit error on one limiting step. It does not control the independent end-bending, collar-removal, hyperbolic-deformation, or preparation losses.

6.3 Monotonicity in height

For $D_2 > D_1$, multiply the difference of the weak equations by $f_{D_2} - f_{D_1}$ on finite truncations and pass to infinity using the source end expansions. Boundary integration gives

$$(D_2 - D_1)(\mathcal{F}_{D_2} - \mathcal{F}_{D_1}) = \int_X [\mathcal{A}(\nabla f_{D_2}) - \mathcal{A}(\nabla f_{D_1})] \cdot \nabla(f_{D_2} - f_{D_1}) \, \mathrm{d}V \geq 0. \tag{6.8}$$

Thus exact fluxes are nondecreasing. The products are integrable on the end: the flux difference is $O(r^{-2})$, the gradient difference is $O(r^{-1})$, and the volume is $O(r \, \mathrm{d}r)$. This yields a useful consistency test for successive computed heights. A violation beyond certified error indicates inconsistent solves or traces; agreement alone does not prove correctness.

7 Compatible Ram Panels and a Concrete Certificate

On a fixed conforming potential space write $v_h = \sum c_j \phi_j$ and use moments $m = Ec$ with an invertible, physically scaled matrix E. On a compatible $H(\mathrm{div})$ space use oriented shared face fluxes and the actual metric/Piola transformation. Divergence constraints are retained exactly; $Dq = 0$ as a cell-incidence equation is sufficient only when it enforces every divergence mode of the chosen flux space. Cell averages alone do not make an arbitrary higher-degree vector field divergence free.

On a curved three-dimensional panel, the flux $(n-1)$-form pulls back with its volume density: in vector form $q \circ F = J\widehat{q}/\det J$. This preserves integrated normal flux. The pointwise condition $|q|_g \leq 1$ is a metric condition, so it must be checked after this pushforward. For polynomial fields and polynomial metric bounds, a Bernstein enclosure of $1 - |q|_g^2$ gives a whole-panel certificate. Rational or nonpolynomial geometry requires a separately enclosed positive denominator and approximation remainder.

Proposition 7.1 (Finite compatible transformations do not change the certificate)**.** *An invertible change of potential and flux coordinates which preserves their physical fields, imposed traces, and incidence constraints leaves $\mathcal{G}$, the certified cut flux, and all inequalities in Theorems 5.2 and 6.2 unchanged.*

Proof. Substitution into the reconstructed fields leaves the integrands, physical metric, volume measure, and boundary traces identical. A consistent incidence transformation is conjugation by the relevant synthesis maps, so it preserves zero divergence and shared normal traces. All bounds concern these physical quantities and hence are unchanged. □

This is exact equivalence, not an improvement in approximation order or proof of faster computation. Poorly conditioned moments can make arithmetic worse. The advantage is the ability to organize adaptive moments while preserving the physical admissibility required for a rigorous certificate.

7.1 Manufactured one-dimensional check

On $[0,1]$ with Euclidean metric, set $u(x) = ax$ with $a > 0$ and the corresponding constant flux $q_* = t(a)$. A conforming comparison $v(x) = ax + \epsilon x(1-x)$ has the same endpoints; any constant $q \in [-1,1]$ is equilibrated. Then

$$\mathcal{G}(v,q) = \int_0^1 \Phi(|a + \epsilon(1-2x)|) \, \mathrm{d}x + \Psi(|q|) - qa. \tag{7.1}$$

For $\epsilon = 0$ this is an exact scalar dual Bregman gap and bounds $|q - q_*|^2/4$. With $a = 0$, $q = 0$, and small nonzero ϵ, the primal energy is $O(|\epsilon|^3)$ while $\int |v'|^2 = \epsilon^2/3$. Thus no uniform quadratic gradient estimate can follow from this gap. The example tests the nonlinear certificate and its limitation; it is not an AH black-hole simulation.

8 Four-Flux and Timelike Mass Error Bounds

The scalar linearization at b is

$$DR_b e = \operatorname{div}_b \operatorname{div}_b e - \Delta_b \operatorname{tr}_b e + 2 \operatorname{tr}_b e, \qquad \operatorname{div}_b U_b(V_a, e) = V_a DR_b e. \tag{8.1}$$

If $R_g + 6 = DR_b e + Q(e)$ and an explicit bound $|Q(e)| \le q_e$ is available, define

$$\varepsilon_R = \frac{1}{16\pi} \int_{r>R} V_0 \{ |R_g + 6| + q_e \} \, \mathrm{d}V_b. \tag{8.2}$$

The integrability is a hypothesis, not something certified by a finite outer sphere.

Proposition 8.1 (Simultaneous four-flux enclosure)**.** *Under the stated integrability and flux existence, each finite-radius integral $p_a(R)$ satisfies $|p_a - p_a(R)| \le \varepsilon_R$. If the numerical surface evaluation has an additional error δ_a, then $|p_a - \widehat{p}_a(R)| \le \varepsilon_R + \delta_a$.*

Proof. Integrate the exact divergence identity over $R < r < R'$ and use $|V_a| \le V_0$. Substitution of $DR_b e = R_g + 6 - Q(e)$ and the triangle inequality give the result after $R' \to \infty$. The quadrature term is another triangle inequality. □

For $|\nabla_b^j e| = O(r^{-\tau})$, $0 \le j \le 2$, the quadratic term is $O(r^{-2\tau})$. Its weighted volume tail behaves like $\int_R^\infty r^{2-2\tau} \mathrm{d}r$, requiring $\tau > 3/2$. At equality $\tau = 3/2$ the logarithmic divergence cannot be discarded without a new cancellation.

Suppose $|p_0 - \widehat{p}_0| \le e_0$ and $|\mathbf{p} - \widehat{\mathbf{p}}| \le e_s$. Let

$$E_- = \widehat{p}_0 - e_0, \quad E_+ = \widehat{p}_0 + e_0, \quad P_+ = |\widehat{\mathbf{p}}| + e_s, \quad P_- = \max(0, |\widehat{\mathbf{p}}| - e_s). \tag{8.3}$$

If $E_- > P_+$, then the four-vector is certified future timelike and

$$\sqrt{E_-^2 - P_+^2} \le m_{\mathrm{AH}} \le \sqrt{E_+^2 - P_-^2}. \tag{8.4}$$

This follows by monotonicity in positive energy and momentum magnitude. Near the null cone a small component error can make the test inconclusive. A formally balanced but inaccurately known vector is not a substitute for this sign check.

If $A \in [A_-, A_+]$, then the Penrose deficit lies in

$$[\, m_- - B(A_+),\ m_+ - B(A_-) \,], \tag{8.5}$$

because B is increasing. A positive lower endpoint certifies strict inequality for that data set. An interval containing zero does not prove equality or geometric rigidity. An area inferred only from a single cut is an upper bound, not a lower bound; a calibrated flux provides the complementary direction.

9 A Residual-Controlled Static Deficit

The source's `hw-public-rigidity` assumes a smooth static system (M^3, q, V) with $\operatorname{Hess} V = V(\operatorname{Ric} + 3q)$, $\Delta V = 3V$, $V > 0$ inside, a connected compact nondegenerate horizon $S = \{V = 0\}$, and a bounded domain U between S and a smooth strictly mean-convex outer face Σ. Set $A = \operatorname{Area}(S)$, $\kappa = |\nabla V|_S$, $d_S = 3A + 2\pi\chi(S)$ and $c = 2A/(3d_S)$. The source proves $d_S > 0$ and, for

$$(\Delta_q - 3)v = -1, \qquad v|_S = c, \quad v|_\Sigma = 0, \tag{9.1}$$

the deficit inequality

$$\mathscr{D} := \frac{2}{3}\int_\Sigma \frac{V}{H} - \int_U V - c\kappa A \geq \frac{3}{2}\int_U V|T(v)|^2 \geq 0, \quad T(v) = \operatorname{Hess} v - v(\operatorname{Ric} + 3q) + q/3. \tag{9.2}$$

9.1 Rechecking the exact identity

Put $B = T - q/3$ and $p = \nabla v - v\nabla V/V$. The static equations and (9.1) give $\operatorname{tr} B = -1$, $\operatorname{div} B = 0$, and $\operatorname{div}(Vp) = -V$. Symmetry cancels the apparent V^{-1} derivatives in

$$\operatorname{div}(VT(p, \cdot)^\sharp) = V|T|^2. \tag{9.3}$$

Although p may be singular at the horizon, $Vp = V\nabla v - v\nabla V$ extends smoothly. Integrating with the correct inward/outward orientations, using Gauss–Bonnet, gives

$$E = \tfrac{2}{3}I - K + c\kappa(cd_S - 2A/3), \qquad I = \int_U V + c\kappa A, \quad K = \int_\Sigma VH(\partial_\nu v)^2, \tag{9.4}$$

where $E = \int_U V|T|^2$. With the displayed c, the extra horizon term vanishes. Weighted Cauchy–Schwarz gives $I^2 \leq JK$ for $J = \int_\Sigma V/H$, hence $JE \leq I(2J/3 - I)$ and $J/I \geq 3/2$. This proves (9.2). The argument is not a generic assertion that any sampled Hessian residual measures distance to a black hole.

Theorem 9.1 (Finite-domain static obstruction certificate)**.** *Assume the smooth geometry above and a known Dirichlet estimate*

$$\|e\|_{H^2(U)} \leq C_E \,\|(-\Delta_q + 3)e\|_{L^2(U)} \quad (e \in H^2 \cap H^1_0). \tag{9.5}$$

Let $\widehat{v} \in H^2(U)$ have the exact boundary trace of v, and let $r = (-\Delta_q + 3)\widehat{v} - 1$. If $C_T = \sqrt{\|V\|_\infty}(1 + \|\operatorname{Ric} + 3q\|_\infty)C_E$, with the H^2 norm dominating both the Hessian and scalar L^2 norms, then

$$\left|\left\|\sqrt{V}T(\widehat{v})\right\|_2 - \left\|\sqrt{V}T(v)\right\|_2\right| \leq C_T\,\|r\|_2\,, \tag{9.6}$$

$$\mathscr{D} \geq \tfrac{3}{2}\left[\max\{0, \left\|\sqrt{V}T(\widehat{v})\right\|_2 - C_T\,\|r\|_2\}\right]^2. \tag{9.7}$$

Proof. The difference $e = \widehat{v} - v$ has zero trace and solves $(-\Delta_q + 3)e = r$. The tensor difference is $\operatorname{Hess} e - e(\operatorname{Ric} + 3q)$. Bound its weighted norm by $C_T\,\|r\|_2$ using (9.5); the reverse triangle inequality proves (9.6). Insert its lower bound into (9.2). □

The smooth bounded Dirichlet problem has such an elliptic estimate, but a *computed* numerical certificate needs a justified numerical upper bound on C_E. A finite matrix inverse does not automatically provide it. Nonzero trace error requires a lifting and its own tensor bound. A strictly positive lower bound in (9.7) excludes equality on this domain; a small bound neither proves exact equality nor controls a global geometric distance. No uniform constant is asserted as Σ goes to infinity.

9.2 Why the global static conclusion remains separate

The source's remaining module 11 uses its normalized end expansion to send the static deficit to zero, derives a sphere horizon using Euler characteristic, constructs a critical-point-free quotient foliation, applies finite-domain substatic Heintze–Karcher rigidity, and extends radiality by an ODE. Recovering the original data still requires the earlier adjoint and null-set steps and the later embedding argument. Our finite-domain certificate is compatible with this route, but it does not certify those inputs or turn a small residual into an exact global isometry.

10 The General Coupled Problem Is Not the Stress Problem

The separate source module 07 uses $D_g = 1 + l^2|\mathrm{d}f|^2$, $d = D_g^{-1}$, $w = l\nabla f/\sqrt{D_g}$, $v = |w|^2$, and

$$\chi = \frac{4d}{4+pv}, \quad A_\chi = I - \frac{4+p}{4+pv} w \otimes w, \quad Z = t + \frac{1}{8}\log D_g. \tag{10.1}$$

Here t is the coupled conformal variable, *not* the constitutive function $t(s)$ of Section 4; the source overloads notation between modules. The implicit derivative $\partial Z/\partial t = (4+pv)/4 > 0$ gives a smooth inverse at fixed background and gradient. The tensor is

$$S = K + \frac{l \operatorname{Hess} f}{\sqrt{D_g}} + w \otimes A + A \otimes w, \tag{10.2}$$

where A is the source's spatial lapse covector, not an area. Its trace equation is $\operatorname{tr}_{A_\chi} S = F$. The companion equation is a true divergence equation with

$$V = uA_\chi\{4\nabla Z + K(w,\cdot) + A(w)w\}, \quad \operatorname{div} V = u\Xi. \tag{10.3}$$

The inner boundary carries prescribed trace heights and a specified normal flux; the outer truncation has $f = Z = 0$. These are neither the Dirichlet stress boundary data nor arbitrary homogeneous Neumann conditions.

The trace equation is in nondivergence form. Replacing $A_\chi : \operatorname{Hess} f$ by $\operatorname{div}(A_\chi \nabla f)$ adds $(\operatorname{div} A_\chi)\cdot\nabla f$ and changes the problem. A symmetric positive matrix from a variational surrogate therefore does not certify the source's operator. Rank-one regularity also requires its gradient-aligned structure; arbitrary measurable elliptic systems do not inherit scalar estimates. In the second equation, the source specifically cancels the spatial lapse drift before applying a scalar end transform. An approximation must retain that cancellation and all coefficient derivatives.

The source's full-cut comparison is $\widehat{g} \geq e^{-4\epsilon} g$, hence $A_{\widehat{g}} \geq e^{-4\epsilon} A_g$. Combining its existence, end, and mass-loss statements gives a time-component inequality, followed by the ordered limit $N \to \infty$ with $\epsilon = B_0 \log N/N$ and removal of strict preparation. Our fixed-D constants in Section 6 do not authorize interchanging any of those limits.

11 Literature, Advantages, and Limitations

Public primary-source literature was checked on October 8, 2026. Allen–Bryden–Kazaras–Khuri obtain a suboptimal-constant spacetime Penrose bound in asymptotically flat and hyperboloidal regimes, with an AH Riemannian corollary [5]. Those regimes and constants should not be conflated with the sharp cosmological cubic target here. Ellithy's 2026 result assumes a specified quasi-final-state

condition [6]; it is not an unrestricted substitute for the present source's initial-data construction. The substatic equality theorem of Borghini–Fogagnolo–Pinamonti [7] concerns a finite domain with its stated regularity, connectedness, and substatic hypotheses. It does not independently establish that the source has produced its global static base.

Mechanism	Advantage of the Ram integration here	Cost or limitation
Constitutive equation	Explicit dual; flux feasibility and dual curvature are visible	Saturation creates a singular dual derivative and linear primal growth
Flux saturation	Fixed-geometry C/D error and ordered height consistency	Barrier, compact volume, and cut constants still need certification
Compatible moments	Divergence, orientation, metric, and traces preserved	No automatic approximation-order or runtime improvement
Finite-radius mass	Four simultaneous tail budgets and timelike sign guard	End integrability is an analytic premise; near-null conditioning is poor
Static deficit	Full residual and domain constant give a rigorous local obstruction	No uniform exhaustion constant or quantitative global rigidity
Coupled deformation	Exact distinction between trace and divergence equations	No certified coupled inverse, degree argument, or global regularity replacement

The results are useful because they expose physical constraints and finite-error budgets previously left implicit in a qualitative limiting step. They do not improve the sharp constant in the Penrose inequality, weaken its global geometric hypotheses, or establish a new equality classification. Classical convex duality, coarea, elliptic estimates, and compatible discretization supply the proof mechanisms; the source-specific combination and exact constitutive formulas are the additions assessed here. An exhaustive priority claim is not made.

12 Further Research and Contribution Assessment

The most concrete next step is to choose an explicit prepared exterior, construct the pole barrier with enclosed constants, and solve several finite-height stress problems with an equilibrated flux reconstruction. Validate the gap, full-cut upper area, outer tail, and C/D bound together. A separate project should study the linear-growth variational relaxation in BV, including boundary trace loss and possible concentration; TR250's $p > 1$ Gamma-convergence theorem does not solve that problem. A third project could obtain domain-uniform static estimates or an actual geometric stability theorem from the deficit, with nondegeneracy and gauge fixed. None follows solely by adding more moments.

Extensions under weaker regularity are plausible for the convex-dual certificate: its proof needs the Green identity, integrability, traces, and an admissible weak solution, rather than analyticity. By contrast, the source's global geometric preparation and rigidity assumptions cannot be weakened on that basis. Stronger bounded-gradient and noncritical hypotheses can yield sharper local primal error bounds, but they must be stated as restrictions, not proven automatically by the representation.

Contribution C3/5. The directly proved contributions are Theorem 5.1, Theorem 5.2, the fixed-geometry rate Theorem 6.2, and the source-specific error propagation in (6.7), Proposition 8.1, and Theorem 9.1. The rate is conditional on explicitly identified source stress existence and end-barrier

inputs; the local dual identity is self-contained. No actual nontrivial AH geometry is numerically solved here. These are notable restricted contributions, not C4/C5-level completion or independent verification of the global theorem.

13 Reproducibility and Verification Scope

The accompanying exact-algebra script checks the constitutive derivative, Legendre equality, stress eigenvalue constraint, universal saturation bound, rational finite-height and mass budgets, compatible moment matrices, and the normalized static-deficit coefficient. Manufactured examples are labeled as such. The analytic proofs establish the continuum statements; finite tests detect transcription errors and check arithmetic only. No sampled test is reported as a proof of ellipticity, end decay, a full-cut infimum, or global rigidity. Source hashes, bounded independent-audit scope, compilation results, and page checks are recorded in the evidence directory and audit manifest. Minor formatting review is limited to two rounds.

References

[1] OpenAI. *The nonmaximal anti-de Sitter Penrose Inequality and original-data rigidity*. October 5, 2026. Pinned commit and exact module locators in Section 1. https://github.com/openai/math.

[2] M. Subbarao. *Compatible Variational Ram Panels*. TR250, September 2026. Private corpus; actual proof locators in Section 3.

[3] M. Subbarao. Astra IE/PDE TR8, `TR8-lin-non-lin.tex`, weak elliptic inverses, domains and Green equivalence. Private corpus.

[4] M. Subbarao. Ram Calculus books, volume 16, `ch05d-clearance.tex`. Polynomial enclosures and exact Bernstein subdivision. Private corpus.

[5] B. Allen, E. Bryden, D. Kazaras and M. Khuri. *Proof of the Spacetime Penrose Inequality With Suboptimal Constant in the Asymptotically Flat and Asymptotically Hyperboloidal Regimes*. 2025. https://arxiv.org/abs/2504.10641.

[6] A. Ellithy. *The spacetime Penrose inequality under a quasi final state hypothesis*. 2026. https://arxiv.org/abs/2605.18730.

[7] S. Borghini, M. Fogagnolo and A. Pinamonti. The equality case in the substatic Heintze–Karcher inequality. *Archive for Rational Mechanics and Analysis* 248 (2024). https://doi.org/10.1007/s00205-024-02022-7; https://arxiv.org/abs/2307.04253.